Huanqi Wen

+8613546387712 | wenhq7@mail2.sysu.edu.cn | huanqiwen.github.io

Education

- **Sun Yat-sen University** — *Guangzhou, China*
  M.S. in Mathematics — 09/2023 - present
  - **GPA:** 3.9/4 (89/100)
  - **Advisor:** Prof. Lei Song
  - **Master's Thesis:** The N_p property of line bundles on toric varieties
- **Sun Yat-sen University** — *Guangzhou, China*
  B.S. in Mathematics, Minor in Economics — 09/2019 - 06/2023
  - **Major GPA:** 3.9/4 (89/100)
  - **Undergraduate Thesis:** On the Minkowski sum of lattice points in lattice polytopes

Research interests

- **Pure mathematics:** toric geometry; tropical geometry; syzygies.

Publications and preprints

- Lei Song, **Huanqi Wen**, Zhixian Zhu. (2024). **On Covering Simplices by Dilations in Dimensions 3 and 4**, arXiv preprint, arXiv:2404.02495

Research experience

- **Positivity of Line Bundles on Toric Varieties** — *09/2023 - present*
  Supervised by Prof. Lei Song
  - **Problem:** motivated by Mukai's conjecture, study whether the pair (X, L) of a projective toric variety X and an ample line bundle corresponding to a polytope P has the N_p properties.
  - **Topics:** stability of syzygy bundles, derived category, resolutions of toric subvarieties, Klyachko's classification of toric vector bundles.
- **Minkowski Sum of Lattice Points in Lattice Polytopes** — *09/2022 - 09/2023*
  Supervised by Prof. Lei Song
  - **Problem:** motivated by Oda's conjecture, study which lattice polytopes are normal, specifically whether the lattice points in an $n+1$ dilation of a polytope P can be written as the sum of a lattice point in P and a lattice point in an n dilation of P.
  - **Results:** By covering simplices with dilations, I proved that simplices with large lattice length in dimensions 3 and 4 are normal; I also wrote computer programs (Python, MATLAB) to verify the covering property and normality of lattice polytopes.

Awards and Scholarships

- **Graduate Scholarship** — *2023, 2024, 2025* ***three times***
  Sun Yat-sen University
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** — *2022*
  Organized by the Consortium for Mathematics and Its Applications (COMAP)
- **Undergraduate Scholarship** — *2020, 2021, 2022* ***three times***
  Sun Yat-sen University

Academic talks

- **Graded Syzygies** *2025 Spring*
  Syzygies Seminar, Sun Yat-sen University [notes]
- **A Series of Lectures on Toric Varieties and Relevant Classical Papers** *2024 Spring - 2024 Fall*
  Toric Geometry Seminar, Sun Yat-sen University [notes]
- **Introduction to Schemes** *2024 Spring - 2025 Spring*
  Algebraic Geometry Seminar, Sun Yat-sen University

Conferences participation

- **2025 Summer Research Institute in Algebraic Geometry**, Colorado State University, USA *07/2025*
- **Stable Reduction of Algebraic Curves (Mini-course)**, Xiamen University, China *05/2025*
- **China Southeastern Algebraic Geometry Symposium VII**, Sun Yat-sen University, China *12/2024*
- **2024 International Summer School of Algebraic Geometry**, Fudan University *07/2024*

Teaching experiences

- **Graduate Teaching Assistant of Linear Algebra**, Sun Yat-sen University *2025 Fall*

Other Experience

- **International Scientific Research and Development Training** *2025 Spring*
  Completed the course on "Constructive Mathematics and Theoretical Computer Science," taught by Professor Vladimir Cherno.
- **Guangdong Digital Talent Training Program** *07/2022 - 08/2022*
  Conducted practical training and internships at China Telecom.
- **Department Student Union Member and Head of a Division** *09/2019 - 06/2021*
  Planned and led diverse student activities within the department

Skills

- Languages: English (IELTS: 7.0), Mandarin (native)
- Programming: C++, Python
- Mathematical tools: MATLAB, SageMath, Macaulay2, LaTeX